

Hwarim (Harley) Hyun

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AI Safety Evaluations Engineer

Agent Evaluation Design | Statistical Experiment Design | Production Evaluation Infrastructure

Systems-minded AI safety evaluation engineer with five years of experience evaluating and optimizing international AI systems: from a 50-billion-vector search engine to all AI safety frameworks at a 8.7B USD/year corporation.

Engineering Work Experience

NAVER

01.2022 - Current

The largest internet service company of Korea, with core business areas including Search, AI, online advertising, commerce, contents, cloud, and fintech. 8.69 billion USD total yearly revenue (as of 2025)

AI Safety Evaluation Engineer (AI Safety Center) - Seongnam, Korea

05.2025 - Current

Create and run safety evaluations for every new NAVER AI product as well as evaluate all product-policy guardrails (11 total NAVER AI agents and services). Run 5-6 safety evaluations per month, checking 10,000–20,000 adversarial queries per test for harmful responses.

- As the second member of the AI Safety Center, designed roughly $\frac{1}{3}$ of NAVER's safety management procedures for AI products from zero, which are now standard procedure for all AI evaluations
- Built an internal automation platform that cut evaluation time by 80% while increasing query volume 5× and risk categories tested 2.2× (24 → 52)
- Built a verification system that raised AI evaluator precision from 73.8% to 85.3%, using 7,800 human labels across 52 risk categories split into tuning and hold-out sets
- Created the automated safety testing process currently under ISO 42001 (AI Management System) audit by LRQA (international certification body) to be the world's first officially certified AI search product
- Awarded an equity grant in 2026, given to less than 10% of all NAVER employees

Software Engineer (Search Engine Team) - Seongnam, Korea

09.2023 - 05.2025

Optimized NAVER's 20-year in-house search engine (10M+ searches a day, index of 50 billion+ vectors). Brought neural retrieval into production under a hard constraint: better quality and more documents on the same hardware, with no model retraining.

- Optimized the space complexity of the engine's grouped-deduplication algorithm from $O(N)$ to $O(K)$, thereby increasing throughput +4.2% at CPU saturation (delivered 6 months earlier than requested, deployed in commerce search)
- Cut vector-search response time 50% by reusing intermediate scores to approximate final scores, while holding search quality at 95% of the full score computation
- Cut index size 50% by implementing an IVF-HNSW index in-house that outperformed Faiss (META's open source indexing system) on build and retrieval time
- Earned top performance ratings in 2023 and 2024 (top rating given to less than 5% of all NAVER employees); selected to speak at DAN24, NAVER's largest engineering conference (800+ engineers), authoring the deck end to end

Software Engineer (ANN Platform) - Seongnam, Korea**01.2022 - 08.2023**

Built and ran the vector search infrastructure behind NAVER's image and video search system (ANN Platform), serving up to 10,000 queries per second across the platform.

- Ran 32 index systems across Korean and Japanese data centers feeding image-to-image and video searches, with the largest index peaking at 6,000 queries per second with no service interruption
- Cut the time to launch a new vector search service from 5 engineer-days to half a day by consolidating indexing that had required manual running across 7 machines into one automated system
- Cut index training time 80% and the update cycle from 7 days to 3, letting new search models reach users in half the time

Software Engineer (Intern) - Seongnam, Korea**01.2021 - 02.2021**

Built the internal management system for NAVER's image-to-image search service during a two-month winter internship. Returned to the same team as a full-time engineer.

Other AI/Engineering Experiences

Co-author, ACM CHI 2026 (Top-tier HCI Venue) - Barcelona, Spain**04.2026**

- "Can LLMs Persuade Humans with Deception?" - a taxonomy of LLM deceptive persuasion strategies with a large-scale empirical study

Singapore AI Safety Red Team Challenge, IMDA Singapore - Singapore**01.2026**

- One of 5 participants representing Korea; contribution featured in NAVER's AI Safety Progress Report and the Korea AI Safety Institute Report
- Invited to review IMDA's published report after participating

Committer, Apache Nemo (Apache Software Foundation)**09.2021 - current**

- Distributed dataflow optimization framework running Apache Beam on YARN and Mesos
- ASF committer status granted by the project committee for sustained contribution

Google Summer of Code 2021 (alumnus)**07.2021 - 08.2021**

- Selected and funded by Google to contribute to Apache Nemo; led to ASF committer status

Co-author, USENIX ATC 2021 (Top-tier systems venue)**07.2021**

- "Refurbish Your Training Data" - faster DNN training by reusing partially augmented samples

First-author, ACM CHI 2019 Workshop (Asian CHI Symposium)**05.2019**

- "CarolTree" - a music education game for children; first of six authors. Built the game and ran a qualitative user study with child participants

Education

Seoul National University - Seoul, Korea**03.2017 - 02.2022**

Bachelor's Degree – Computer Science and Engineering, Statistics (double major)

Skills

Evaluation: LLM/agent evaluation design, red-teaming, adversarial dataset construction, benchmark design

Engineering: Python, evaluation harnesses, Spark pipelines, statistical experiment design